

# Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports

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**Abstract.** Operational flood streams contain duplicate, incomplete, and adversarial reports, while clustering and priority scores are often evaluated without measuring their scheduling consequences. We audit a fail-closed batch pipeline in which observable reports form a sparse graph, unresolved reports go to review, duplicate families feed a bounded score, and evaluator-only truth is joined after predicted-cluster scheduling. The clustering benchmark contains 80 synthetic runs from generator 3.0.0: additive Louvain has mean ARI 0.9191, product Leiden 0.9165, and product Louvain 0.9072. Product minus independently tuned additive is  $-0.0118$  (bootstrap 95% CI  $[-0.0226, -0.0024]$ ), but its Holm-adjusted test is not significant ( $p = .2560$ ). A matched-density comparison reverses the point estimate but is also inconclusive. In a separate candidate-4.1 bundle, the revised score has NDCG@5 0.6669 versus 0.6763 for a random comparator and does not show a general alignment advantage; exact duplicates are invariant but coordinated high-confidence campaigns remain a failure mode. Dispatch results similarly show no general benefit over the legacy score and a clear cost relative to nearest-first. These are synthetic, within-generator diagnostics: boundedness and clustering accuracy do not establish source authenticity, policy validity, field benefit, or deployment readiness.

**Keywords:** Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

## 1 Introduction and Related Work

During emergencies, crisis streams mix repeated reports of one event with nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency-message research addresses volume, verification, and event extraction [10]. CrisisMMD and FloodNet capture useful disaster modalities, although FloodNet is an aerial post-flood scene-understanding dataset rather than a multimodal report stream [1,9]. These resources do not jointly provide physical-incident linkage, rescue priority, and field dispatch outcomes.

*Social sensing and event extraction.* Social-media users can act as real-time sensors [10], but the usual endpoint is event detection or classification rather than the consequences of sending a field resource to a predicted incident.

*Consolidation and clustering.* Graph community detection with Louvain or Leiden [3,14], and density methods such as DBSCAN and HDBSCAN [4] offer plausible consolidation mechanisms. Multiplicative spatial-range terms occur in bilateral filtering [13]; we adapt that structure to a product graph, while testing the exact gate and its bounds here.

*Humanitarian logistics and dispatch.* Relief-routing models expose competing equity, efficacy, and efficiency objectives [5]. We use this perspective to motivate a bounded priority ranking, but boundedness is not an expert-endorsed policy. A key gap is propagation of false splits, merges, and noise into a scheduler that cannot consult incident truth.

We address these gaps with a fail-closed batch pipeline: reports with observable location and time enter a sparse graph, incomplete reports go to manual review, duplicate families feed a bounded ranking, and evaluator-only truth joins after predicted-cluster scheduling. We ask: (RQ1) does independently selected product composition improve clustering over additive composition? (RQ2) does the duplicate-aware score align with an independently generated benefit under report-level stress tests? (RQ3) how do predicted split, merge, and noise errors propagate into dispatch? The questions use two explicitly separated artifact bundles.

The contributions are (i) an observable-only pipeline with explicit review and evaluator boundaries; (ii) auditable threshold statements and a proposed duplicate-aware score; and (iii) a paired diagnostic reporting clustering, ranking, dispatch, and adverse results. The evidence remains synthetic and within-generator, so the artifact is not a validated rescue policy.

## 2 Observable Pipeline and Methods

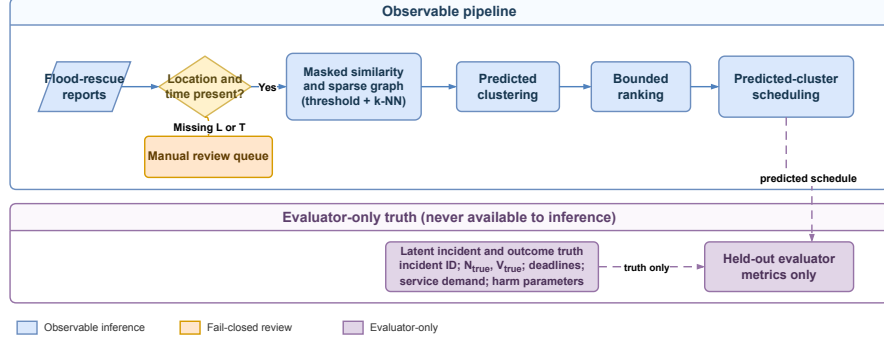
Figure 1 fixes the information boundary. Observable report fields flow through graph construction, ranking, and a batch scheduler. Missing location or timestamp routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

### 2.1 Report Attributes and Missingness

Each report is represented by the compact tuple

$$r_i = (L_i, T_i, F_i, E_i, N_i, V_i, Q_i),$$

where  $L_i$  is a GPS location and  $T_i$  is the observable report timestamp (“created\_at”). Flood evidence  $F_i$  and urgency evidence  $E_i$  are in  $[0, 1]$ ;  $N_i \in \mathbb{Z}_{\geq 0}$  is the reported trapped-person count;  $V_i \geq 0$  is a vulnerability-evidence value; and  $Q_i \in [0, 1]$  is a provenance/confidence heuristic.  $Q_i$  is not a calibrated probability of truth. The implementation also carries observable support fields such as image presence, source type, province, and missing-field flags; these support  $Q_i$  and duplicate fingerprints but are not additional latent labels.



**Fig. 1.** Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

The score  $Q_i$  is computed from observable image presence and a capped count  $n_i^{\text{corrob}}$  of nearby corroborating payloads using the proposed heuristic

$$Q_i = \text{sigmoid}(b_0 + b_1 \mathbf{1}\{\text{image}_i\} + b_2 \log(1 + n_i^{\text{corrob}})),$$

$$(b_0, b_1, b_2) = (-.2, 1.4, .9).$$

Here  $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$ . This is a pipeline rule, not a calibrated truth model. No independent receipt timestamp or source family is assumed. The admission indicator  $a_i = \mathbf{1}\{L_i, T_i \text{ observed}\}$ , the masks  $I_F(i, j), I_E(i, j)$ , and  $m_{ij} = I_F(i, j) + I_E(i, j)$  are derived pairwise quantities; hence  $a_i a_j = 0$  prevents an edge whose geographic or temporal separation is undefined. Missing  $F$  or  $E$  is not zero-imputed. Following the general missing-variable kernel problem [11], we use the fail-closed construction

$$C_{ij} = \frac{m_{ij}}{2} \exp\left[-I_F(i, j) \frac{|F_i - F_j|}{\tau_F} - I_E(i, j) \frac{|E_i - E_j|}{\tau_E}\right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0.$$

Thus complete context gives the usual exponential similarity, one shared field gives at most one half, and no shared field gives no inferred agreement. In particular, when  $m_{ij} = 2$ ,  $C_{ij} = \exp[-|F_i - F_j|/\tau_F - |E_i - E_j|/\tau_E]$ . All RQ1 reports contain  $F$  and  $E$ , so the evaluated RQ1 graph uses the complete-data reduction; missingness is a specified, not an empirically claimed, extension.

## 2.2 Similarity Graph and Derived Bounds

For Haversine distance  $d_{ij}$  and time difference  $\Delta t_{ij}$ , define the Gaussian affinity [8] and normalized temporal decay [16]

$$\begin{aligned} G_{ij} &= \exp[-d_{ij}^2/(2\sigma^2)], & T_{ij} &= \exp[-\Delta t_{ij}/\tau_t], \\ w_{ij}^\times &= a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij}), & w_{ij}^+ &= a_i a_j (\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \end{aligned} \quad (1)$$

The exponential form is a precedent, not a claim that the complete missingness-aware expression is copied from prior work [2]. The product is a spatial gate motivated by spatial-range products such as bilateral filtering [13]; both compositions and all weights here are defined by this paper.

The implementation uses a common spatial candidate pool, an empirical nonzero-weight quantile, strict above-threshold edges, endpoint top- $k$  retention, and Louvain or Leiden. Product and additive candidates are tuned independently; their comparison is therefore a pipeline comparison. All scales are positive and all composition weights are nonnegative.

**Theorem 1 (Product edge localization).** *Let  $B = \beta + \gamma > 0$ . If a product edge with threshold  $0 < \theta < B$  is retained, then  $d_{ij} < \sigma\sqrt{2\log(B/\theta)}$ . For  $\theta \geq B$  the strict retained set is empty.*

*Proof.* Because  $a_i a_j \leq 1$  and  $T_{ij}, C_{ij} \leq 1$ ,  $w_{ij}^\times \leq B \exp[-d_{ij}^2/(2\sigma^2)]$ . Combining this bound with  $\theta < w_{ij}^\times$  and taking logarithms gives the result.

**Corollary 1 (Conditional component bound).** *If a retained product component has observed unweighted hop diameter  $h < \infty$ , then its geographic diameter is less than  $h\sigma\sqrt{2\log(B/\theta)}$ .*

This is a conditional post-hoc statement: the pipeline does not control  $h$ . For the additive form, a finite geographic cutoff exists only when  $B < \theta < B + \alpha$ , with radius  $\sigma\sqrt{2\log(\alpha/(\theta - B))}$ . These are deductions from the weight definitions, not empirical performance guarantees.

### 2.3 Duplicate-Aware Bounded Ranking

Exact duplicate payloads share a canonical fingerprint. Near duplicates are joined by deterministic connected components under the observable envelope; this transitive rule is not complete linkage. Let  $g$  index the resulting evidence families. For the current candidate source snapshot, the revised estimator uses confidence-gated family maxima and avoids treating overlapping reports as disjoint people. Write  $\tilde{N}_i = \min(N_i, N_{\text{ref}})$  and  $\tilde{V}_i = \min(V_i, V_{\text{cap}})$  for the nonnegative clipped claims:

$$\begin{aligned} \bar{E} &= \frac{\sum_g \max_{i \in g} Q_i E_i}{\max(1, \sum_g \max_{i \in g} Q_i)}, & \bar{F} &= \max_{g, i \in g} Q_i F_i, \\ \bar{N} &= \min\left(1, \frac{\log(1 + \max_{g, i \in g} Q_i \tilde{N}_i)}{\log(1 + N_{\text{ref}})}\right), & \bar{V} &= \max_{g, i \in g} Q_i \tilde{V}_i, \\ P &= [1 + (\mu - 1) \tanh(\bar{V}/s)] (\omega_E \bar{E} + \omega_F \bar{F} + \omega_N \bar{N}). \end{aligned} \tag{2}$$

The candidate source snapshot associated with the RQ2/RQ3 artifacts specifies  $\omega = (.34, .33, .33)$ ,  $\mu = 2$ ,  $s = 10$ ,  $N_{\text{ref}} = 500$ , and  $V_{\text{cap}} = 50$ ; the Colab configuration audit must confirm these values against the locked bundle before submission. The formula is a proposed policy heuristic: maxima limit duplicate

multiplicity, the logarithm compresses heavy-tailed demand, clipping bounds individual evidence, and the hyperbolic tangent saturates vulnerability amplification. Reliability-aware truth discovery motivates using  $Q_i$  but does not derive these maxima [15]. Transformation, normalization, and weights require sensitivity and expert validation [12,5].

### 3 Experimental Design

*Two artifact bundles.* RQ1 uses generator 3.0.0, seeds 1–80, and 20 development, 20 calibration, and 40 benchmark-test runs. The last 40 are a benchmark split within the same generator regime, not an unseen OOD test. RQ2 and RQ3 use candidate generator 4.1.0, test seeds 3000–3039, and bundle SHA `a7c7a3f7...f2d7e1`; their manifests are stored with the result CSVs. Both bundles are geographically anchored to the Copernicus EMSR848 Central Viet Nam flood activation; incidents, reports, duplicates, campaigns, and labels remain synthetic.

*RQ1 clustering.* The calibration notebook selects each composition independently by mean ARI, then standard deviation and canonical configuration JSON. Product uses  $\sigma = 700$  m,  $\tau_t = 60$  min, quantile .90, and  $k = 8$ ; additive uses  $\alpha = 1$ , the same scales and  $k$ , and quantile .95. The common candidate pool and grid make the comparison reproducible but not causal. ARI is primary; NMI, pair F1, diameter, edge fraction, and operational-noise fields are descriptive. Paired differences use 10,000 bootstrap resamples, Wilcoxon tests, and Holm correction [6]. K-means is the coordinate comparator [7].

*RQ2 score isolation.* The evaluator generates an independent relevance signal from latent incident benefit and computes NDCG@5, Spearman correlation, and top-five overlap. It uses oracle incident grouping only to isolate score behavior; inference sees observable reports only. Exact duplicates, near duplicates, contradictions, missingness, and coordinated campaigns are injected as stress scenarios.

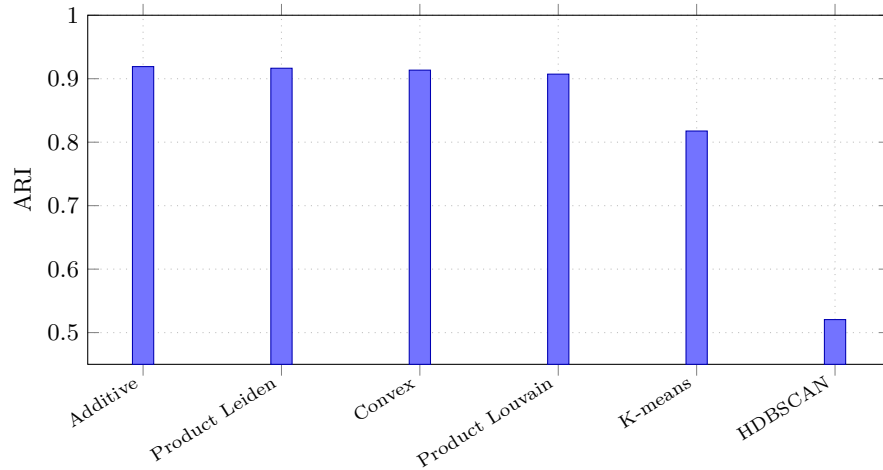
*RQ3 error propagation.* The scheduler receives Product, matched-density Additive, or oracle partitions and compares revised, legacy, FIFO, nearest-first, equity-aging, and workload smoothing policies. The conservative evaluator permits one destination per incident, charges split trips and fake-only dispatches, and scores unserved incidents at 480 minutes. Paired effects use the CSV column `mean`, whose positive direction is already oriented as better for the candidate; raw lower-is-better differences have the opposite sign.

### 4 Results

Additive Louvain has the highest mean ARI. Product minus independently tuned additive is  $-0.01184$  with CI  $[-0.02257, -0.00240]$ , 9 wins, 19 ties, and 12 losses; raw  $p = .0987$  and Holm  $p = .2560$ . Product minus matched-density additive is

**Table 1.** RQ1 benchmark-test summary (40 runs). Intervals are bootstrap 95% CI low-high; diameter is descriptive.

| Method             | ARI [95% CI]        | Pair F1 | Mean diam. (km) |
|--------------------|---------------------|---------|-----------------|
| Additive Louvain   | .9191 [.8995,.9381] | .9247   | .3477           |
| Product Leiden     | .9165 [.8979,.9351] | .9223   | .3547           |
| Convex Louvain     | .9135 [.8940,.9320] | .9195   | .3547           |
| Product Louvain    | .9072 [.8864,.9271] | .9138   | .3585           |
| Coordinate K-means | .8176 [.7976,.8384] | .8292   | .2977           |
| HDBSCAN            | .5205 [.4565,.5811] | .5787   | 1.2527          |

**Fig. 2.** RQ1 benchmark-test ARI means. The plotted values are the same summary used in Table 1; bootstrap intervals remain reported in the table.

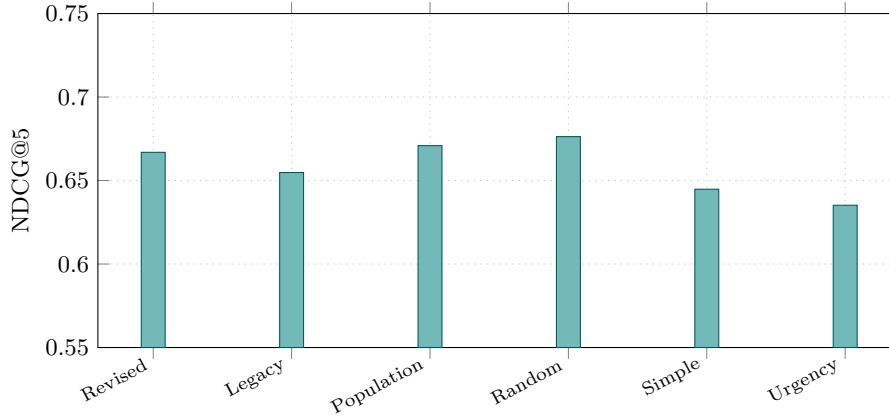
+0.00989 with CI  $[-0.00100, 0.02082]$  and Holm  $p = .2560$ . Product Leiden versus Product Louvain has Holm  $p = .1120$ . Thus neither composition advantage is established. Product and additive have nearly equal retained edge fractions (.03011 and .03021) and mean degrees (9.90 and 9.94), supporting the matched-density interpretation. Product has zero fake absorption and zero fake rejection but 16.49% false destinations: fakes avoid genuine destinations while still forming fake-only destinations, so this is not misinformation robustness.

RQ2 shows no general alignment benefit: revised NDCG@5 is .6669 (CI [.6306,.7018]), below population-only (.6709) and random (.6763), and no comparator contrast is Holm-significant. Revised priority is invariant to  $2\times$ ,  $5\times$ , and  $10\times$  exact duplication. Near duplicates and some missingness scenarios improve over legacy, but a coordinated high-confidence campaign produces drift .1688 and false-priority lift .8828; revised drift is worse than legacy ( $\Delta = -.1379$ , Holm  $p \approx 9.1 \times 10^{-12}$ ). Boundedness is therefore not a guarantee of ranking reliability.

RQ3 reaches the same operational caution. On the Product partition, revised versus legacy has harm effect  $-14.34$  (Holm  $p = .675$ ) and deadline effect  $+.0188$

**Table 2.** Headline RQ2/RQ3 results from the candidate-4.1 artifacts. For paired rows, a positive effect means the revised candidate is better; negative effects are therefore adverse.

| Question | Contrast and metric                 | Effect | Holm $p$              |
|----------|-------------------------------------|--------|-----------------------|
| RQ2      | Revised NDCG@5 (absolute)           | .6669  | –                     |
|          | Random NDCG@5 (absolute)            | .6763  | –                     |
|          | Revised – legacy NDCG@5             | +.0121 | .5882                 |
|          | Exact-duplicate priority drift      | 0      | –                     |
|          | Coordinated-campaign drift          | .1688  | –                     |
| RQ3      | Revised – legacy, Product harm      | –14.34 | .675                  |
|          | Revised – legacy, Product deadline  | +.0188 | .160                  |
|          | Revised – nearest, Product harm     | –69.88 | $1.8 \times 10^{-7}$  |
|          | Revised – nearest, Product deadline | –.1771 | $3.0 \times 10^{-15}$ |
|          | Product – matched Additive harm     | –18.34 | .000151               |

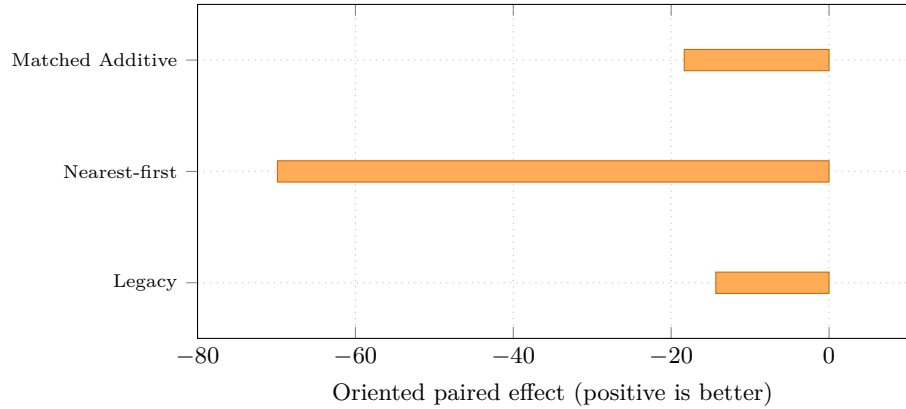


**Fig. 3.** RQ2 alignment comparison across the candidate-4.1 policies. The revised score is not above the random or population-only comparators; bootstrap intervals are retained in the RQ2 summary artifact.

(Holm  $p = .160$ ), so no general improvement is established. Revised is significantly worse than nearest-first on both harm and deadline, while its advantage over FIFO is a comparison with a weak baseline. Product is also worse than oracle under the revised policy and worse than matched-density Additive on harm. Clustering errors thus propagate into dispatch outcomes.

## 5 Discussion and Threats to Validity

The results separate mathematical safeguards from empirical utility. Product gating supplies a finite edge bound in its positive-threshold domain, but the independently tuned product pipeline does not outperform additive clustering. The revised score removes exact-duplicate inflation, yet its ranking is not better than simple comparators in general and can amplify coordinated high-confidence



**Fig. 4.** RQ3 Product-partition harm effects for the revised policy. Negative values indicate higher harm than the comparator; the values are taken from the oriented paired-comparison results.

campaigns. Dispatch outcomes confirm that a bounded score cannot substitute for a validated operational objective.

All reports, incidents, and relevance signals are synthetic. RQ1 varies seeds within generator 3.0.0; RQ2/RQ3 use a separate candidate-4.1 bundle. Neither tests mechanism-shift OOD transfer or field performance. RQ1 does not provide an empirical missingness test because all evaluated reports contain the four core attributes. No expert panel validated weights, caps, or utility, and no result measures extraction accuracy, device latency, memory, energy, provenance authenticity, or model accuracy. A prospective mobile deployment is therefore only an architecture proposal.

## 6 Artifact and Conclusion

The result folders contain the RQ1 summary and paired comparisons, RQ2/RQ3 CSVs, and SHA manifests for the candidate bundle. The manuscript treats these CSVs as the numerical source of truth; the obsolete `paper/short_results.tex` and generated snapshots are not imported. Before submission, the Colab audit must record the candidate source files, priority configuration, and bundle hash, and every number in this paper must trace to a current CSV row.

In conclusion, the study establishes conditional graph bounds and documents competitive within-generator clustering, exact-duplicate invariance, and several important negative results. It does not establish ranking alignment, dispatch benefit, misinformation robustness, policy validity, or deployment readiness. The contribution is an auditable stress-test boundary around those claims, not a validated rescue policy.



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